Human-Computer Interaction (HCI)

Mids Notes | Chapter 1 se 9 | Aasan Roman Urdu mein, examples ke saath, har topic ke neeche paper wala English jawab

Pehle: Is data mein kya hai?

Aap ki zip file mein **9 PowerPoint slide files** hain. Yeh Alan Dix, Janet Finlay, Gregory Abowd aur Russell Beale ki mashhoor kitab **'Human-Computer Interaction' (3rd Edition)** ke lecture slides hain, kitab ke **pehle 9 chapters**. (File ke naam 'e3-chap-xx' ka matlab edition 3, chapter xx.)

File	Kya hai	Mashhoor topics
Chapter 1: The Human	Insaan: dekhna, sunna, chhoona, harkat; memory (sensory, short-term, long-term); sochna (reasoning, problem solving); ghaltiyan; emotion; logon mein farq.	Fitts' law, 7±2 chunks, slips vs mistakes
Chapter 2: The Computer	Computer ke hisse: input devices (keyboard, mouse, touch), output (screens, printers), VR, sensors, memory, processing speed, networks.	QWERTY, CRT vs LCD, RAM vs disk, Moore's law
Chapter 3: The Interaction	User aur system ki baat cheet: Norman ka model, Abowd-Beale framework, ergonomics, interaction styles (command line, menu, WIMP...), experience aur value.	Gulf of execution/evaluation, WIMP
Chapter 4: Paradigms	HCI ki tareekh: batch processing se ubiquitous computing tak, direct manipulation, hypertext, WWW.	Time-sharing, WIMP, metaphor, ubiquitous computing
Chapter 5: Interaction Design Basics	Design kaise karte hain: users ko jaanna (persona), scenarios, navigation, screen layout, affordance, colour, prototyping.	Persona, breadcrumbs, grouping/alignment
Chapter 6: HCI in the Software Process	Software lifecycle mein HCI, usability engineering, ISO 9241, prototyping, design rationale.	Waterfall, usability specification, QOC/IBIS
Chapter 7: Design Rules	Usability ke principles, standards, guidelines, Shneiderman ke 8 golden rules, Norman ke 7 principles, design patterns.	Learnability, flexibility, robustness
Chapter 8: Implementation Support	Interface banane ke tools: windowing systems, toolkits, UIMS, MVC aur PAC models.	Event loop, MVC, Seeheim
Chapter 9: Evaluation Techniques	Design ko check karna: cognitive walkthrough, heuristic evaluation, experiments, think aloud, interviews, questionnaires, eye tracking.	Variables, hypothesis, think-aloud

Kaun si cheezein is data mein nahi hain: kitab ke chapters 10 se 21 (jaise Universal design, Cognitive models, Task analysis, Dialog notations) is zip mein nahi hain, aur slides mein kuch jagah unka zikr aata hai ('see chapter 10, 12...'). Slides kaafi purani hain (PDA, CRT monitor, 2002 ke numbers), is liye maine jahan zaroori hua wahan **aaj ki misaalein** bhi di hain.

Kaise parhna hai: har topic mein pehle **asaan baat** (misaal ke saath), phir **peela box** (example), aur aakhir mein **neela box** jis mein chhote English jumle hain jo paper mein seedha likh sakte ho. Isi neele box wale jawab ek alag **short English PDF** mein bhi hain.

Chapter 1: The Human (insaan)

1.1 HCI mein 'Human' ko kyun samajhte hain?

Agar aap ko ek jootay banana ho to pehle paon ka naap lena parta hai. Isi tarah computer ka interface banane se pehle **insaan ko samajhna** zaroori hai: woh kaise dekhta, sunta, yaad rakhta aur sochta hai. Insaan ek **information processing system** ki tarah hai: maloomat andar aati hai (input), yaad hoti hai (memory), soch kar kaam ki jati hai (processing), aur kaam kiya jata hai (output).

Paper mein aise likho:
The human is an information-processing system. Information comes in through the senses (vision, hearing, touch), is stored in memory (sensory, short-term, long-term), is processed by reasoning and problem solving, and results in actions. Emotion and individual differences also affect performance.

1.2 Vision (dekhna)

Dekhne ke **2 marhale** hain: (1) aankh se roshni ka **physical reception**, (2) dimagh mein us ki **processing aur matlab nikalna**.

Aankh kaise kaam karti hai: roshni cheez se reflect ho kar aankh mein aati hai, aur retina par **ulti (upside-down)** tasveer banti hai. Retina mein do qisam ke cells hain: **Rods** (kam roshni mein dekhne ke liye, andhere mein) aur **Cones** (rang dekhne ke liye). **Ganglion cells** pattern aur harkat pehchante hain. Isay camera ki tarah socho: lens, aur film (retina) jis par tasveer banti hai.

Size aur depth: Visual angle batata hai ke koi cheez hamari nazar ka kitna hissa ghere hue hai (cheez ke size aur door hone se related). **Visual acuity** baareek tafseel dekhne ki salahiyat hai (limited). Hum jaani-pehchani cheez ka size **constant** samajhte hain chahe door se chhoti dikhe (aap ka dost door se chhota dikhta hai magar aap usay chhota nahi samajhte). **Overlapping** (ek cheez ka doosre ke aage hona) se depth ka andaza hota hai.

Brightness aur Colour: Brightness roshni ke level par hamara subjective reaction hai. Roshni zyada ho to acuity bhi barhti hai aur flicker bhi. **Colour** ke 3 hisse: **hue** (kaunsa rang), **intensity** (kitna tez), **saturation** (kitna gehra/pakka). Cones rang ke wavelengths ko mehsoos karte hain. **Blue ki acuity sab se kam hai. 8% mardon aur 1% auraton** mein colour blindness hoti hai.

Visual system compensate karta hai harkat aur roshni ki tabdeeli ko, aur context se confusion door karta hai. Kabhi kabhi zyada compensation se **optical illusions** hote hain (Ponzo illusion aur Muller-Lyer illusion: do barabar lambai ki lakeerein alag lambai ki lagti hain).

Reading (parhna): pehle shakal nazar aati hai, phir dimagh us ko zubaan ke hisaab se decode karta hai, phir syntax, semantics aur pragmatics se matlab nikalta hai. Aankh parhte waqt **saccades** (tezi se chhalang) aur **fixations** (theher kar dekhna) karti hai; **samajh fixations ke dauran hoti hai**. Lafz ki **shakal (word shape)** pehchan mein madadgar hai. Computer screen par **negative contrast** (kala text safed background par) parhne mein behtar hai.

Example: Isi liye interface mein khaas baat par **blue ka istemal na karo** (blue acuity kam hai), aur colour-blind logon ke liye sirf rang par bharosa na karo (lal/hara ke saath icon ya text bhi do).

Paper mein aise likho:
Vision has two stages: physical reception of the stimulus and processing/interpretation of it. The retina has rods (low light) and cones (colour). Colour has hue, intensity and saturation. Blue acuity is lowest, and about 8% of males and 1% of females are colour blind. Reading involves saccades and fixations, and perception happens during fixations.

1.3 Hearing (sunna)

Awaaz humein mahaul ke baare mein batati hai: kitni door, kis taraf. Kaan ke **3 hisse**: **outer ear** (awaaz ko bachata aur amplify karta hai), **middle ear** (awaaz ki lehron ko vibrations bana kar andar bhejta hai), **inner ear** (chemical signals ban kar auditory nerve se dimagh tak). Awaaz ki 3 khasiyatein: **pitch** (frequency, bareek ya bhaari), **loudness** (amplitude, tez ya dheemi), **timbre** (qisam ya quality, jaise gitar aur piano ki awaaz mein farq).

Insaan **20 Hz se 15 kHz** tak sun sakta hai. Kaan shor mein se kaam ki awaaz **chhaant** leta hai: ise **cocktail party phenomenon** kehte hain (bheed mein sirf apna naam sun lena).

Paper mein aise likho:
Hearing provides information about distance, direction and objects. The ear has an outer, middle and inner part. Sound has pitch (frequency), loudness (amplitude) and timbre (quality). Humans hear about 20 Hz to 15 kHz, and can focus on one sound over background noise (the cocktail party phenomenon).

1.4 Touch (chhoona)

Touch mahaul ka feedback deta hai aur nabeena logon ke liye bohat zaroori hai. Jild mein 3 qisam ke receptors: **thermoreceptors** (garmi/sardi), **nociceptors** (dard), **mechanoreceptors** (dabao). Ungliyan zyada sensitive hain. **Kinesthesis** jism ki position ka ehsaas hai jo comfort aur performance par asar karta hai.

Paper mein aise likho:
Touch gives feedback via skin receptors: thermoreceptors (heat/cold), nociceptors (pain) and mechanoreceptors (pressure). It is especially important for visually impaired users. Kinesthesis is the awareness of body position.

1.5 Movement aur Fitts' Law

Kisi cheez par response dene ka waqt = **reaction time + movement time**. Reaction time stimulus par depend karta hai: **visual ~200 ms, auditory ~150 ms, pain ~700 ms**. Movement time umar aur fitness par depend karta hai.

Fitts' Law batata hai screen par kisi target tak pahunchne mein kitna waqt lagta hai:

Mt = a + b log2(D/S + 1)

Yahan **Mt** = movement time, **D** = faasla (distance), **S** = target ka size, **a aur b** tajarbe se nikle hue constants. Matlab: target **jitna bara aur jitna qareeb**, utna jaldi pahunch sakte ho. **Design usool:** targets bade rakho, faasla kam rakho.

Example: Mobile par 'Send' ka button bada aur angootay ke qareeb hona chahiye, chhota aur door nahi. Isi liye important buttons bade bante hain.

Paper mein aise likho:
Response time is reaction time plus movement time. Fitts' Law: Mt = a + b log2(D/S + 1), where Mt is movement time, D is distance, S is target size, and a and b are constants. Targets should be as large and as close as possible.

1.6 Memory (yaad dasht)

Memory ki **3 qismein** hain, bilkul school ke ghar ki tarah: pehle **sensory memory** (darwaza: bohat thori der ke liye cheezein pakadta hai), phir **short-term memory** (kaam ki meiz: abhi ka kaam), phir **long-term memory** (almari: hamesha ke liye). Kaun si cheez andar jayegi, yeh **attention** aur **rehearsal** (dohrana) par depend karta hai.

Memory	Matlab	Example
Sensory memory	Sense se aane wali cheezon ka buffer. **Iconic** (nazar), **echoic** (awaaz), **haptic** (touch). Musalsal naye se overwrite hoti rehti hai.	Sparkler se andhere mein goli ka nishaan banana.
Short-term memory (STM)	Temporary 'scratch pad'. Access ~70 ms, bohat jaldi mit'ti (~200 ms), capacity **7 ± 2 chunks**.	Phone number ko dial karne tak yaad rakhna.
Long-term memory (LTM)	Hamari saari maloomat. Access ~1/10 second, decay bohat kam, capacity bohat zyada.	Apna ghar ka pata, school ke din.

7 ± 2 chunks ka matlab: hum ek waqt mein sirf 5 se 9 cheezein yaad rakh sakte hain. Cheezon ko **chunks (gutthon)** mein baant do to zyada yaad hota hai. Slide ke examples: **212348278493202** (bohat lamba, yaad rakhna mushkil), **0121 414 2626** (gutthon mein, aasan), **HEC ATR ANU PTH ETR EET** (bekaar lafz, mushkil) jo asal mein **THE CAT RAN UP THE TREE** ke harf hain, yaani meaning ho to aasan hota hai.

LTM ki do qismein: Episodic (waqiaat ki tarteeb: 'kal main ne kya kiya') aur **Semantic** (hawalon aur ilm ka structure: 'Paris France ka capital hai'). Semantic LTM episodic se bani hoti hai.

LTM ke models (ilm kaise rakha hota hai):

Model	Matlab	Example
Semantic network	Cheezein ek jaal (network) ki tarah rishton se juri hain. Child node parent ki khasiyat **inherit** karta hai, is se inference hota hai.	Canary ek bird hai, bird ke par hote hain, to canary ke par hain.
Frames	Maloomat ke dhanche (slots) jin mein value bharte hain: fixed, default, variable. Type-subtype rishte.	DOG: legs=4 (fixed), sound=bark (default), colour (variable). COLLIE DOG ka subtype.
Scripts	Kisi hamesha ki tarah wali soorat-e-haal ka model: entry conditions, result, props, roles, scenes, tracks.	'Vet ke paas jana': kutta beemar, vet khula; props: meiz, dawai; roles: vet, owner; scenes: reception, intezar, muaina, payment.
Production rules	**IF condition THEN action** wale usool; tareeqa-e-kaar (procedural) ilm.	IF kutta dum hila raha hai THEN thapki do. IF ghurra raha hai THEN bhaag jao.

LTM mein kaise jata hai: Rehearsal (dohrane) se STM se LTM mein. **Total time hypothesis:** jitna zyada waqt dohrao utna zyada yaad. **Distribution of practice:** sabaq ko **waqt ke saath phaila kar** seekhna behtar hai (ek raat mein ratta ke bajaye roz thora thora). Cheez agar **structure, matlab aur jani-pehchani** ho to yaad karna aasan hai.

Bhoolna (Forgetting): Decay (maloomat bohat aahista mit'ti hain) aur **Interference** (nayi cheez purani ko hata deti hai = retroactive; purani nayi mein ruqawat bani = proactive). Yaadasht **selective** hai aur **emotion** se asar leti hai.

Retrieval (wapas nikalna): Recall (khud yaad kar ke nikalna; cues madad karte hain, jaise categories ya tasveer) aur **Recognition** (dekh kar pehchanna 'haan yeh pehle dekha tha'). **Recognition recall se aasan hai**, kyunke maloomat khud cue ban jati hai. Isi liye menu (options nazar aate hain) command line (khud yaad karna) se aasan hota hai.

Paper mein aise likho:
There are three memory types: sensory (iconic, echoic, haptic), short-term (limited to 7 ± 2 chunks, fast decay) and long-term (huge capacity, episodic and semantic). LTM models: semantic networks, frames,

scripts and production rules. Rehearsal moves information to LTM; forgetting is due to decay and interference. Recognition is easier than recall.

1.7 Thinking (sochna): Reasoning

Sochne ke 3 tareeqe:

Qisam	Matlab	Example
Deduction (nateeja nikalna)	Diye hue bayanat (premises) se **lazmi** nateeja nikalna.	Agar Friday ho to wo kaam par jayegi. Aaj Friday hai. To wo kaam par jayegi. (Note: logically sahi nateeja sach hi ho yeh zaroori nahi: 'agar baarish ho to zameen khushk hai; baarish hai; to zameen khushk hai' logically sahi dhancha hai magar sach nahi.)
Induction (aam bana dena)	Dekhe hue maamlon se **nadekhe maamlon** ke baare mein aam qanoon banana. Kamzor hai: sirf ghalat sabit ho sakta hai, sahi nahi. Magar kaam ka hai.	Humne jitne haathi dekhe sab ki soond thi, to har haathi ki soond hoti hai.
Abduction (nateeje se wajah)	Waqiya dekh kar **wajah** ka andaza lagana. Ghalat wajah bhi nikal sakti hai.	Sam sharaab pi kar tez gari chalata hai. Maine Sam ko tez chalate dekha, to wo nashe mein hoga.

Wason's cards: 4 cards: **7, E, 4, K**. Usool: 'Agar kisi card par ek taraf vowel (aeiou) hai to doosri taraf even number hai.' Is ko sach/jhoot check karne ke liye kaun se cards palatne chahiye? **Jawab: E aur 7.** E palat kar dekho (peeche even hona chahiye). 7 palat kar dekho (agar peeche vowel nikla to usool ghalat). 4 aur K ko palatne ki zaroorat nahi. Aksar log E aur 4 chunte hain, jo ghalat hai, kyunke insaan **negative evidence** ko achi tarah istemal nahi karte.

Paper mein aise likho:
Deduction derives a logically necessary conclusion from premises. Induction generalizes from seen cases to unseen cases (it can only be disproved, never proved). Abduction reasons from an event to its likely cause and can give false explanations. Humans are poor at using negative evidence (Wason's cards).

1.8 Problem solving, Errors aur Mental models

Problem solving anjaan kaam ka hal dhoondna hai, apni maloomat se. Teen nazariye: **Gestalt** (hal 'achanak samajh' aur masle ko naye sire se dekhne se milta hai: productive aur reproductive), **Problem space theory** (masle ki halatein/states hoti hain; hum 'legal operators' se aik halat se doosri mein jate hain, aur **heuristics** jaise **means-ends analysis** use karte hain), aur **Analogy** (naye masle ka hal purane milte jultay masle se nikalna; agar dono domains bohat alag hon to mushkil). **Skill acquisition:** maahir log maloomat ko **chunk** kar lete hain taake STM par bojh kam ho.

Ghaltiyon ki qismein:

Qisam	Matlab	Example
Slip	Iraada **sahi** tha magar karte waqt **ghalat** ho gaya (be-dhyani, kamzor jismani maharat).	Aap ne 'Save' dabana tha magar 'Delete' par click ho gaya. Hal: behtar interface design.
Mistake	Iraada hi **ghalat** tha (samajh ghalat).	Aap ne socha ke 'Delete' dabane se file Recycle Bin mein jayegi, magar wo hamesha ke liye gayi. Hal: system ki behtar samajh.

Mental model: insaan system ke baare mein apne dimagh mein ek **naqsha** bana leta hai ke yeh kaise kaam karta hai. Agar yeh naqsha asal system se alag ho to **ghaltiyan** hoti hain.

Paper mein aise likho:
Problem solving theories include Gestalt, problem space theory (states, operators, heuristics such as means-ends analysis) and analogy. Skilled users chunk information. A slip is a correct intention carried out wrongly; a mistake is a wrong intention caused by incorrect understanding. A wrong mental model of the system leads to errors.

1.9 Emotion (jazbaat)

Emotion kaise kaam karta hai is par teen nazariye: **James-Lange:** emotion hamari jismani reaction ki **tashreeh** hai. **Cannon:** emotion stimulus par ek **psychological reaction** hai. **Schacter-Singer:** emotion hamari jismani reaction ko **poore halaat ki roshni mein parakhne** ka nateeja hai. Emotion mein dimaghi aur jismani dono reactions shamil hain.

Jismani stimulus par biological reaction ko **affect** kehte hain. **Positive affect** creative problem solving mein madad deta hai; **negative affect** sochne ko tang kar deta hai. Donald Norman: 'negative affect aasaan kaam bhi mushkil bana deta hai, positive affect mushkil kaam aasaan kar deta hai.'

Design par asar: stress se problem solving mushkil hoti hai; **relaxed users** design ki kamiyon ko maaf kar dete hain; **khoobsurat aur khushgawar interface** positive affect barhata hai.

Paper mein aise likho:
Emotion involves both cognitive and physical responses. Affect is the biological response to stimuli: positive affect helps creative problem solving, negative affect narrows thinking. For design, stress makes problem solving harder, relaxed users forgive design flaws, and pleasing interfaces increase positive affect.

1.10 Individual differences aur Psychology ka istemal

Har insaan alag hai. Farq **lambay arse** ke hain (jins, jismani aur zehni salahiyatein), **chhote arse** ke (stress aur thakawat) aur **badalte rehne wale** (umar). Design se pehle poochho: **kya mera faisla aabadi ke kisi hisse ko bahar to nahi kar dega?** (jaise bujurg ya colour-blind).

Psychology ka seedha istemal: blue ki acuity kam hai, is liye zaroori tafseel ke liye blue nahi. Magar sahi istemal ke liye context aur experiment ki shartein samajhna zaroori hai. Is ka nichor guidelines (chapter 7), cognitive models aur evaluation techniques (chapter 9) mein jama hai.

Paper mein aise likho:
People differ in long-term (abilities), short-term (stress, fatigue) and changing (age) ways. A designer should ask whether a design decision will exclude part of the user population. Psychology knowledge is distilled in guidelines, cognitive models and evaluation techniques.

Chapter 2: The Computer

2.1 Computer system ke hisse

Interaction ko samajhne ke liye computer ke hisse samajhna zaroori hain, kyunke **devices tay karte hain ke interaction ka andaz kya hoga**. Computer system 7 hisson se bana hai: **input devices** (text entry aur pointing), **output devices** (screens aur digital paper), **virtual reality** (khaas devices), **physical interaction** (awaaz, haptic, bio-sensing), **paper** (print aur scan), **memory** (RAM aur permanent), **processing** (raftar aur networks).

Interactivity: pehle **batch processing** tha: punched cards tayyar karo, lambi intezar, line printer ka output, aur ghalti ho to dobara. Ab zyada tar computing **interactive** hai: foran feedback, user control

mein. Aur ab computers har jagah hain: ghar mein (TV, microwave, washing machine), jeb mein (phone, smart card, car key, USB). Sawal: **kya tez hona hamesha behtar hai?** (nahi: jaise help screen itni tez scroll ho ke parha na jaye).

Paper mein aise likho:
A computer system consists of input devices, output devices, virtual reality devices, physical interaction (sound, haptic, bio-sensing), paper, memory and processing/networks. The devices dictate the styles of interaction a system can support.

2.2 Text entry devices

Device	Kaam / khasiyat	Example
Keyboard (QWERTY)	Sab se aam. Keypress se character code jata hai. **QWERTY layout typing ke liye behtareen nahi**, asal mein typewriter ke jam hone se bachane ke liye banayi gayi thi. Alternatives hain magar log QWERTY ke aadi hain.	Normal keyboard
Alphabetic keyboard	Keys A-B-C ki tarteeb mein. Na trained, na beginners ke liye tez.	
Dvorak	Aam huroof dominant ungliyon ke neeche, haath badal badal kar. **10-15% tez**, thakawat kam. Phir bhi market mein nahi chala.	
Special keyboards	RSI (thakawat) kam karne ke liye ya ek haath ke liye.	Maltron left-handed keyboard
Chord keyboard	Sirf 4-5 keys; huroof keys ke **combination** se. Chhota, jaldi seekha jata hai. Nuqsan: samaji mukhalifat, thakawat.	Wearables ke liye niche
Phone pad aur T9	Numeric keys ko baar baar dabana (2 = a b c). **T9** predictive: dictionary se lafz ka andaza.	hello = 43556 (T9)
Handwriting recognition	Pen aur tablet se likhna. Masle: joined writing ko alag huroof mein torna, alag handwriting.	PDA, tablet
Speech recognition	Awaaz se text. Behtar jab ek user, limited vocabulary. Masle: shor, uchaaran, bari vocabulary, alag bolne wale.	Voice assistant
Numeric keypad	Number tezi se. Calculator aur phone ka layout **alag** hai (1-2-3 upar ya neeche).	ATM

Paper mein aise likho:
Text entry devices include keyboards (QWERTY, Dvorak, alphabetic, special, chord), phone pads with T9 predictive entry, handwriting recognition and speech recognition. The QWERTY layout is not optimal for typing but is kept because of the large social base of QWERTY typists.

2.3 Positioning, pointing aur drawing devices

Device	Kaam / khasiyat	Example
Mouse	Haath mein pakra; table par chalta hai; screen cursor ko hilata hai. **Indirect device** (haath table par, nishan screen par). Relative movement. Mechanical (ball) ya optical (LED, dhool se kam asar).	Desktop computer
Touchpad	Chhota touch tablet; ungli phera kar cursor chalate hain. 'Acceleration' settings zaroori: tez phera = door tak, aahista = sahi position.	Laptop
Trackball / thumbwheel	Ulta mouse: ball ghoomti hai, housing sthir. Gaming mein tez. Thumbwheel: CAD (X-Y dial) ya scroll.	
Joystick / keyboard nipple	Danda dabane ka dabao = raftar. Games, jahaz. Nipple: laptop keyboard ke beech mein mini joystick.	
Touch screen	Ungli ya stylus ki maujoodgi pehchanta hai. **Direct** device. Fayde: tez, kisi khaas pointer ki zaroorat nahi, menu selection ke liye achha, gande mahaul mein sahi. Nuqsan: screen par nishan, ungli mota (sahi nahi), chhoti jagah mushkil, bazu thak jata hai.	Phone, ATM
Stylus aur light pen	Pen jaisa pointer, screen par seedha likhna. Bohat direct, magar screen chhupa sakta hai. Light pen ab shayad hi.	Tablet
Digitizing tablet	Cross-hair wala mouse jaisa device khaas surface par. Bohat accurate (maps).	
Eyegaze	Aankh ki taraf se control: menu ki cheez ko dekh kar chuno. Hands-free. Headset se accuracy.	Evaluation, accessibility
Cursor keys	Chaar keys (upar, neeche, left, right); sasti magar dheemi. Aam: ulta 'T'.	
Discrete controls	Phone aur TV ke cursor pads/mini joysticks; menu selection ke liye.	TV remote

Direct vs indirect: Direct mein aap seedha cheez ko chhoote ho (touch screen). **Indirect** mein device alag aur nishan alag (mouse, trackball, joystick).

Paper mein aise likho:
Pointing devices include the mouse, touchpad, trackball, joystick, touch screen, stylus, digitizing tablet, eyegaze and cursor keys. A mouse is an indirect device because hand movement on the desk moves a cursor on the screen. A touch screen is a direct device: it is fast and needs no special pointer, but is imprecise and tiring for the arm.

2.4 Display devices (screens)

Bitmap screen: screen bohat saare rangeen **dots (pixels)** ka majmua hai. **Resolution** = pixels ki tadaad (jaise 1024x768) ya density (dpi, 72-96). **Aspect ratio** = chauraai aur lambai ka nisbat (4:3 ya 16:9). **Colour depth** = har pixel ke kitne rang (black/white, 256, ya 8 bits per R/G/B = millions of colours). **Anti-aliasing:** tirchi lakeeron ke 'jaggies' (seedhiyaan jaisi) ko line ke rang ke shades se naram karna, text par bhi.

Display	Matlab	Khaas baat
CRT (Cathode Ray Tube)	Electron gun se electrons phosphor wali screen par; screen chamakti hai. Purane TV aur monitors.	**Sehat ke khatre:** X-rays (mostly absorb), radio frequency, electrostatic field (rashes), electromagnetic fields (aankhon ke masle).

Display	Matlab	Khaas baat
LCD (Liquid Crystal Display)	Chhota, halka, **radiation ka masla nahi**. Roshni reflect hoti hai, emit nahi, is liye aankhon par kam zor. Voltage se crystal ka polarisation aur rang badalta hai.	Laptop, phone, digital watch
Random scan / vector display	Seedhi lakeerein draw; jaggies nahi; musalsal redraw. Khaas instruments mein.	
DVST	Random scan jaisa magar persistent (flicker nahi); selectively mita nahi sakte.	Analogue oscilloscope
Large displays	Plasma, video walls, projectors (haath screen ko chhupa sakta hai), back-projected.	Lecture hall
Situated displays	**Public jagah** par; maloomat us jagah se related. **Location matters.**	Hermes: office ke darwaze par chhoti screen, jis par notes chhore jate hain.
Digital paper	Patli, mudne wali sheets, electronically update, magar kagaz jaisi dikhti hain (chhote gole ya coloured liquid).	E-ink reader

Sehat ke asaan usool: screen ke bohat qareeb na baitho; bohat chhote fonts na use karo; lambay arse tak bina break ke na dekho; screen ko tez khidki ke saamne na rakho; roshan mahaul mein kaam karo; posture aur ergonomics ka dhyan rakho.

Paper mein aise likho:
A bitmap display is made of many coloured pixels; resolution, aspect ratio and colour depth describe its quality. Anti-aliasing softens jagged edges. CRTs have health hazards (radiation, fields), while LCDs are lighter with no radiation problems. Other displays: large, situated (location-specific) and digital paper.

2.5 Virtual Reality aur 3D

VR mein user 3D jagah mein harkat karta, pakadta aur 3D dekhta hai. Cheezein: **cockpit aur virtual controls**; **3D mouse** (6 degrees of freedom: x, y, z + **roll, pitch, yaw**); **data glove** (fibre optics se ungli ki position); **VR helmets** (sar ki harkat aur nazar); **whole-body tracking**. **Desktop VR** mein aam screen aur mouse se perspective se 3D ka ehsaas. **VR headset** mein har aankh ke liye alag chhoti screen, thori alag angle se, 3D asar.

VR motion sickness: sar hilao to display mein **delay (lag)** aati hai; aur aankhon ka angle (depth) aur focus mein **conflict** hota hai; conflicting cues se chakkar aate hain. **Simulators aur VR caves:** deewaron par scene, asal controls, hydraulic rams.

Paper mein aise likho:
Virtual reality uses 3D positioning devices (3D mouse with six degrees of freedom: x, y, z, roll, pitch, yaw), data gloves, helmets and body tracking. VR motion sickness comes from time delay (lag) and conflicting depth cues between eye angle and focus.

2.6 Physical controls, sensors aur sounds

Dedicated displays: analogue (dials, gauges, lights) aur digital (LCD, LED); **head-up displays** (jahaz ke cockpit mein zaroori maloomat shishe par). **Sounds:** beeps aur clicks error aur confirmation ke liye (key click). **Haptic devices:** touch aur feeling: games mein vibration aur force feedback, simulation mein surgery ka ehsaas. Example: **BMW iDrive** menu items ke liye chhoti 'bumps' deta hai taake bina dekhe chun sakein. **Physical controls** context ke mutabiq hote hain: bohat saaf karne wale smooth buttons, bade buttons, saaf dials.

Environment aur bio-sensing: hamare aas paas sensors hain: gari ki light ka door switch, ultrasound detectors, RFID tags, temperature, weight, location; aur hamare jism ke: iris scanner, body temperature, heart rate, galvanic skin response, blink rate.

Paper mein aise likho:
Dedicated displays can be analogue or digital; head-up displays show key information in the user's line of sight. Sound gives feedback and warnings, and haptic devices give touch feedback (for example vibration, BMW iDrive). Sensors detect the environment and the user's body (RFID, heart rate, iris scan).

2.7 Paper: printing aur scanning

Printing: tasveer chhote dots se banti hai. Ahem baatein: **resolution (dpi)**, **speed (pages per minute)**, **cost**. **Dot-matrix** (ribbon aur pins, 80-120 dpi), **ink-jet** (siyahi ke chhote qatray, 300+ dpi), **laser** (photocopier ki tarah; toner; 600+ dpi), **thermal** (garam hone wala khaas kagaz; shop tills).

Fonts: font text ka andaz hai, size **points** (1 pt = 1/72 inch) mein. **Pitch:** fixed-pitch (har harf barabar chauda, Courier) ya variable (i aur m alag chaude, Times). **Serif** (siron par ubhre hue hisse: Times, Palatino) ya **Sans-serif** (seedhe sire: Helvetica). **Readability:** chhote huroof (lowercase) lafz ki shakal ke liye behtar; **UPPERCASE** alag huroof aur ghair-lafz ke liye (flight number BA793). Serif lambi chhapi hui lines ke liye achha, magar **screen par sans-serif aksar behtar**.

Page Description Languages (PostScript): poora page bitmap bana kar bhejna bohat bara hota hai; is ke bajaye page ka **description** bhejte hain (lines, curves, text). **WYSIWYG** (jo dekho wohi chhape) lekin screen (72 dpi, landscape) aur print (600+ dpi, portrait) hamesha thore alag hote hain.

Scanners aur OCR: Scanner kagaz ko bitmap bana deta hai (flat-bed ya hand-held; 600-2400 dpi). **OCR (Optical Character Recognition)** bitmap ko wapas text banata hai. Masle: alag fonts, page ka format (columns, tasveerein). **Paper-based interaction:** kagaz sirf output nahi, input bhi ho sakta hai (Xerox PaperWorks glyphs, khaas pen jo likhne ki jagah jaanta hai).

Paper mein aise likho:
Printers form images from dots (dot-matrix, ink-jet, laser, thermal), measured in dpi and pages per minute. Fonts differ in size (points), pitch (fixed or variable) and serif or sans-serif. Page description languages such as PostScript describe a page instead of sending a bitmap. Scanners convert paper to bitmaps and OCR converts bitmaps back into text.

2.8 Memory aur storage

Memory	Matlab	Example
RAM	Silicon chips; ~100 ns access; **volatile** (bijli band to mit jati hai); ~100 MB/s transfer. Short-term memory jaisa kaam.	Chal rahe programs ka temporary kaam
Magnetic disks	Floppy (~1.4 MB), hard disk (40 GB se 100s GB); access ~10 ms. Long-term storage.	
Optical disks	Lasers se parhna/likhna; magnetic se zyada mazboot. CD-ROM, DVD.	
Flash memory	Silicon based magar **persistent**. PDA, camera, USB.	USB drive

Virtual memory: bohat programs ho aur RAM kam ho to kuch programs **temporary disk par** rakh dete hain, RAM bara dikhne lagti hai; magar **swapping** se computer **dheema** ho jata hai. **Compression:** storage kam karna. **Lossless** (bilkul wapas milta hai: ZIP, GIF; jaise AAAAAAAAAABBBBBCCCCCCCC = 10A5B8C) aur **lossy** (kuch kho jata hai magar insaan ko pata nahi chalta: JPEG, MP3). **Storage formats:** text: ASCII, UTF-8, RTF, SGML, XML; images/audio/video: JPEG, GIF, MPEG, WAV waghera. **Access:** bare store mein **index** se dhoondte hain; **jo index karoge wohi dhoond sakoge**; forgiving

systems jaise SOUNDEX (McCloud ~ MacCleod).

Paper mein aise likho:
RAM is fast, volatile short-term memory; disks (magnetic, optical) are long-term storage; flash memory is persistent silicon memory. Virtual memory stores some programs temporarily on disk but swapping slows the system. Compression is lossless (ZIP, GIF) or lossy (JPEG, MP3).

2.9 Processing aur networks

Finite processing speed: designer aksar maan lete hain ke processor bohat tez hai aur interface ko pechida bana dete hain. Phir masle: cursor kisi cheez ke aage nikal jata hai (keypresses buffer hone ki wajah se); 'icon wars' (icon par click, kuch nahi, doosra click, phir sab windows ek saath khul jati hain). Bohat tez hona bhi bura ho sakta hai (help screen itni tez scroll ho ke parhi na jaye). **Moore's Law** (Gordon Moore, 1965): processor ki raftar har **18 mahine mein double** hoti hai (memory har ~12 mahine mein). **'Infinitely fast machine' ek myth hai**: asal machine ke hisaab se design karo.

Interactive performance ki limitations: Computation bound (hisaab mein der), **Storage channel bound** (disk se memory mein data ki rukawat), **Graphics bound** (display update mehnga), **Network capacity** (network ki raftar). **Networked computing:** bari memory aur processing, dusre log (email, groupware), shared resources (web). Issues: **delay, conflicts, unpredictability**. **Internet:** 1969 DARPANET (4 sites) se; common protocols: **TCP** (packets, reliable channel) aur **IP**; email aur HTTP inhi par bane.

Paper mein aise likho:
Designers should not assume an infinitely fast machine. Moore's law says processor speed doubles roughly every 18 months. Interactive performance is limited by computation, storage, graphics and network capacity. Networks give access to large resources and other people but cause delays, conflicts and unpredictability.

Chapter 3: The Interaction

3.1 Interaction kya hai? Domain, Goal, Task

Interaction ka seedha matlab **user aur system ke beech baat cheet (communication)** hai. Lekin sirf itna nahi: iska ek saamaji aur tanzeemi (organizational) pehlu bhi hai. Teen terms jo aksar poochay jate hain:

Term	Matlab	Example
Domain	Woh **kaam ka maidan** jis par hum kaam kar rahe hain.	Graphic design
Goal	Hum **kya hasil karna** chahte hain.	Ek thos lal triangle banana
Task	Us goal tak **kaise** pahunchte hain (operations/actions).	Fill tool chuno, triangle par click karo

Paper mein aise likho:
Interaction is communication between the user and the system. Domain is the area of work, a goal is what the user wants to achieve, and a task is how the goal is achieved in terms of actions.

3.2 Donald Norman ka Interaction Model (7 stages)

Norman batata hai ke user jab computer se kaam karta hai to dimagh mein **7 qadam** chalte hain. Yeh **user ke nazariye** se hai. Qadam do hisson mein hain: pehle **Execution** (kaam karna), phir **Evaluation** (nateeja dekhna).

Qadam	Matlab	Example
1. Establish the goal	Goal tay karna.	Mujhe ek photo email karni hai.
2. Formulate the intention	Iraada banana.	Mail app mein attachment lagaunga.
3. Specify the action sequence	Kaunse actions kis tarteeb mein.	Compose dabao, paperclip dabao, photo chuno, Send.
4. Execute the action	Asal mein karna.	Click/tap karna.
5. Perceive the system state	System ki halat dekhna.	Screen par 'Sent' dikhta hai.
6. Interpret the system state	Us ka matlab nikalna.	Yani mail chali gayi.
7. Evaluate against the goal	Goal se milana.	Haan, photo bhej di, kaam mukammal.

Do 'gulf' (khaiyan): Gulf of Execution: user jo karna chahta hai aur system jo karne ki ijazat deta hai, in mein **farq**. (Aap ko pata nahi ke photo kaise attach hoti hai.) **Gulf of Evaluation:** user jo nateeja **umeed** karta hai aur system jo asal mein **dikhata** hai, in mein farq. (Aap ne Send dabaya magar pata nahi chalta ke gayi ya nahi.) **Behtar interface in khaiyon ko kam karta hai.**

Human error: slip aur mistake: Slip: sahi goal aur sahi iraada magar karne mein ghalti (hal: behtar interface design). **Mistake:** goal hi ghalat samjha (hal: system ki behtar samajh).

Paper mein aise likho:
Norman's model has seven stages: establish the goal, formulate the intention, specify the action sequence, execute the action, perceive the system state, interpret it, and evaluate it against the goal. The gulf of execution is the difference between the user's intended actions and the actions the system allows. The gulf of evaluation is the difference between what the user expects and what the system shows.

3.3 Abowd aur Beale ka Interaction Framework

Yeh Norman ka extension hai. Is mein **4 hisse** hain: **User (U)**, **Input (I)**, **System (S)**, **Output (O)**. Har hissa apni alag **zubaan** bolta hai (User ki task language, Input ki input language, System ki core language, Output ki output language). **Interaction = in zubaanon ke beech tarjuma (translation)**, aur **interaction ke masle = tarjume ke masle**.

Flow: user ka iraada (task) -> input ke actions mein tarjuma -> system ki halat (core) mein tabdeeli -> output (display) mein dikhna -> user us ki tashreeh karta hai. Yeh sirf computers ke liye nahi, sab interaction ke liye aam framework hai, aur systems ka **muqabla** karne mein madad deta hai.

Paper mein aise likho:
The interaction framework (Abowd and Beale) has four parts: User, Input, System and Output, each with its own language. Interaction is translation between these languages, and problems in interaction are problems in translation. The user's intention is translated into input actions, then into a change of system state, shown as output, and interpreted by the user.

3.4 Ergonomics aur Industrial interfaces

Ergonomics (human factors): interaction ki **jismani khasiyaton** ka mutala. Standards aur guidelines banane mein achha. Misalein: **controls aur displays ki tarteeb** (function ya istemal ke hisab se group), **mahaul** (baithne ka intezam sab size ke logon ke liye), **sehat** (position, temperature, roshni, shor), **rang ka istemal** (lal = khatra, hara = theek, colour-blindness ka khayal).

Industrial vs office interface: office mein data **textual**, tabdeeli **slow**, mahaul **saaf**. Industry mein data **numeric**, tabdeeli **fast**, mahaul **ganda** ('oil-soaked mouse'). **Glass interface:** purane dials/knobs ki jagah screens aur keypads: sasta, flexible, kayi tarah ke display, sahi values; magar cheez ki jagah se juda, context ka nuqsan, pechida. **Indirect manipulation:** office mein user nakli (artificial) duniya se kaam karta hai (direct), industry mein asal duniya se **interface ke zariye**; masle: feedback aur delay.

Paper mein aise likho:
Ergonomics studies the physical characteristics of interaction (arrangement of controls and displays, environment, health issues, use of colour). Industrial interfaces differ from office interfaces: data is numeric, changes are fast and the environment is harsh. Glass interfaces replace dials with screens; industrial work often involves indirect manipulation of the real world.

3.5 Interaction styles (interface ke andaaz)

Style	Matlab	Example
Command line	Instructions seedha likhna (function keys, chhote lafz, poore lafz). Repetitive kaam aur **experts** ke liye achha; seedha access. Naam mayne wale hon.	Unix
Menus	Options ki list screen par. Options nazar aate hain: **recognition** (kam yaad karna). Selection: number, harf, arrow, mouse. Sensible grouping zaroori.	ATM menu, phone settings
Natural language	Jaani-pehchani zubaan (bol kar ya likh kar). Masle: mubham (vague), do matlab, theek karna mushkil. Hal: subset samajhna, key words.	Voice assistant
Query / question-answer	User ko sawalon se guzaara jata hai; novices ke liye, magar limited. Query language (SQL): expert ke liye.	Information kiosk, SQL
Form-fill	Kagaz ke form jaisi screen; data entry ke liye. Achhe design aur ghalti sudhaarne ki sahulat chahiye.	Registration form
Spreadsheet	Cells ka grid; value ya formula; formula dusre cells par depend. Form-fill ki jadeed shakal. (VisiCalc, Lotus 1-2-3, MS Excel)	Excel
WIMP	**W**indows, **I**cons, **M**enus, **P**ointers. PCs ka default andaaz.	Windows, macOS
Point and click	Bas kisi cheez par click (icon, link, map). Kam typing.	Web browser, multimedia
3D interfaces	Virtual reality, 3D workspaces (extra jagah, roshni aur parchhai se depth). Behad istemal confusing.	3D desktop

Paper mein aise likho:
Common interaction styles: command line, menus, natural language, question/answer and query dialogue, form-fills and spreadsheets, WIMP, point-and-click and three-dimensional interfaces. Menus rely on recognition rather than recall, so they are easier for novices, while command lines suit experts.

3.6 WIMP ke elements

Element	Matlab	Example
Windows	Screen ke aise hisse jo alag alag behave karte hain. Text ya graphics, move/resize, overlap ya tiled. **Scrollbars** aur **title bars**.	
Icons	Chhoti tasveer jo object, window ya action ko dikhati hai. Window ko **iconise** (chhota) kar sakte hain.	
Pointers	Pointing aur selecting par bharosa. Mouse, trackpad, joystick, trackball, cursor keys.	
Menus	Operations ki list. **Masla:** screen ki bohat jagah. **Hal:** pop-up.	
Buttons	Alag, chhote hisse jinhein dabane se action. **Radio buttons** (ek hi choice) aur **check boxes** (kayi choices).	
Toolbars	Icons ki lambi line, aam kaamon tak jaldi pahunch. Customizable.	
Palettes aur tear-off menus	Menu jo chahiye tab maujood nahi hota; palette chhoti window hai actions ki, ya menu 'tear off' ho kar palette ban jata hai.	Drawing app ke shapes
Dialogue boxes	Popup windows jo ahem event batati hain ya maloomat maangti hain.	File save karte waqt filename box

Menus ki qismein: Menu bar (upar): **pull-down** (mouse pakad kar neeche), **drop-down** (click par khulta), **fall-down** (sirf mouse le jane se). **Contextual / pop-up menus** (selected object ke actions jahan aap ho wahin). **Pie menus** (gol: target bada, har option ka faasla barabar; magar aam nahi). **Cascading menus** (menu ke andar menu). **Keyboard accelerators** (Ctrl + letter).

Paper mein aise likho:
WIMP stands for windows, icons, menus and pointers. Windows are independent areas of the screen, icons are small pictures representing objects or actions, menus offer choices, and pointers select items. Related elements are buttons (radio buttons for exclusive choices, check boxes for non-exclusive), toolbars, palettes and dialogue boxes.

3.7 Interactivity: look, feel, initiative, error

Look and feel: sab WIMP systems ke elements wohi hote hain magar alag systems (macOS aur Windows) **alag behave** karte hain. **Look + behaviour = look and feel**. **Initiative:** purane question-answer mein computer ke paas initiative tha; WIMP mein **user** ke paas. Exception: **modal dialog boxes** (aati hain aur khud nahi jaati: ghaltiyon aur zaroori qadam ke liye theek, magar ehtiyaat se).

Speech-driven interfaces tez hote ja rahe hain magar ghair-sahi; mazboot dialogue ke liye system apni samajh wapas bolta hai ('aap New York se Boston ka ticket chahte hain?'). **Error and repair:** ghaltiyon se bacha nahi ja sakta magar unhein **pakarna aur theek karna** aasan hona chahiye (airline booking ke misaal mein user ke 'no' par system ek ek cheez confirm karwata hai).

Context: interaction par **saamaji aur tanzeemi** mahaul asar karta hai: dusre log (impress karne ki khwahish, muqabla, nakami ka dar), **motivation** (dar, wafadari, ambition), aur nakara systems se tang aa jana.

Paper mein aise likho:
Look and feel is appearance plus behaviour; different window systems with the same elements behave differently. In WIMP systems the user has the initiative, except for modal dialogue boxes. Errors cannot always be avoided, so systems should make errors easy to detect and repair. Social and organizational context also affects interaction.

3.8 Experience, engagement aur fun

Ghar, entertainment aur shopping mein sirf **use kar sakna** kaafi nahi, log usay **use karna chahein** bhi. **Flow** (Csikszentmihalyi): dilchaspi tab hoti hai jab kaam na bohat mushkil (anxiety) na bohat aasan (boredom). **Zone of proximal development** (education): woh cheezein jo madad se aap kar sakte ho.

Example: Real cracker vs virtual cracker: Asli cracker mein bura mazaq, plastic khilauna, kagaz ki topi; kheench kar 'bang'. Virtual mein mazaq, web toy, mask, click kar ke 'bang'. Designer ko **ehsaas** (suspense, surprise, shared experience) bhi design karna hota hai, sirf function nahi.

Paper mein aise likho:
It is not enough that people can use a system; they must want to use it. Flow is the balance between anxiety and boredom. Designers must consider experience elements such as excitement, suspense, surprise and shared experience in addition to function.

3.9 Physical design aur Value

Physical design ki pabandiyan: ergonomic (kam az kam button size), physical (high-voltage switch bare), legal/safety (cooker ke controls oonche), context (saaf karne mein aasan), aesthetic (achha dikhe), economic (mehnga na ho). Yeh aksar **aapas mein takrati** hain, to **trade-offs**. Misal: cooker ka front panel bade ke liye behtar, rear panel bachhe ke liye mehfooz. Ergonomics ke liye bade buttons chahiye magar jagah nahi (MiniDisc remote): hal multifunction controls.

Fluidity: kya bahar ki jismani shakal andar ke logical asar ko dikhati hai? On/off button ki halat nazar aaye; ulti harkat ka ulta asar (arrow buttons, twist controls). **Spring-back controls** (chhor do to wapas): one-shot buttons, joystick; achha: bare selection sets, bura: chhupi halat. **Physical layout:** controls ka **logical rishta = spatial grouping**. **Compliant interaction:** mechanical buttons aur rotary knobs apni halat dikhate hain aur user aur machine dono control kar sakte hain.

Managing value: log koi cheez tab hi use karte hain jab us ki **perceived value** ho **aur** value cost se zyada ho. Value sirf paisa nahi: kaam jaldi hona, mazaa, doosron ka bhala. Cost: download time, paisa, seekhne ki mehnat. **Discounted future:** log mustaqbil ki value aur cost ko bohat kam samajhte hain, isliye seekhne mein jhijak: **kam rukawat aur foran nazar aane wali value** chahiye. **Organisational design:** coercion (hukm), enculturation (company values samjhana), emergence (process aisa ke log khud company ki value ko apni value samjhein). **Aam sabaq:** agar kisi se kaam karwana hai to usay **aasaan** banao aur uski **value** samjho.

Paper mein aise likho:
Physical design involves constraints (ergonomic, physical, legal and safety, context, aesthetic, economic) that often conflict, so trade-offs are needed. People use something only if its perceived value exceeds its cost. Because people heavily discount future value, designs need low barriers and high perceived present value.

Chapter 4: Paradigms (HCI ki tareekh)

4.1 Paradigm kya hota hai?

Paradigm ka matlab hai **sab se mashhoor nazariya ya soch ka dhancha**. Jaise physics mein Newton se Einstein tak soch badli, waise HCI ki tareekh mein bhi **paradigm shifts** aaye: jab nayi technology aayi to insaan aur computer ke rishte ki soch badal gayi. Do sawal: interactive system kaise banaya jaye ke woh **usable** ho, aur us ki usability kaise **naapi jaye**?

Paper mein aise likho:
A paradigm is a predominant theoretical framework or world view. In HCI, a paradigm shift occurs when a new computing technology creates a new perception of the human-computer relationship. Studying HCI

history means studying a series of such shifts.

4.2 Paradigm shifts ka silsila

Paradigm	Matlab	Saal / naam
Batch processing	Shuruati: **impersonal** computing. Cards bana kar jama, lambi intezar, printout.	1940s-50s
Time-sharing	Ek computer **kayi users** ke liye. J.C.R. Licklider (ARPA). Interactive computing.	1960s
Video Display Units	Kagaz ke bajaye screen. **Sutherland ka Sketchpad (1962)**: computer se visualize aur manipulate.	1962
Programming toolkits	**Engelbart** (Stanford Research Institute): 'augmenting man's intellect'; NLS/Augment system demo (1968). Sahi toolkit pechida systems banane ke building blocks deta hai.	1963-68
Personal computing	**Papert ki LOGO** (bachon ke liye graphics programming). **Alan Kay ka Dynabook** (Xerox PARC): har shakhs ka apna chhota taqatwar computer. Jitna aasan, utna taqatwar.	1970s
Window systems aur WIMP	Insaan ek se zyada kaam kar sakta hai; windows 'topic badalne' ke liye. **Xerox Star (1981)** pehla commercial windowing system.	1981
Metaphor	Computing ko asal duniya se jodna: desktop, typing, spreadsheet. Masle: sab kaam metaphor mein fit nahi, cultural bias.	
Direct manipulation	**Shneiderman (1982)**: graphic interaction ki khoobi. **Apple Macintosh (1984)**, WYSIWYG.	1982-84
Language vs action	Action hamesha lafzon se behtar nahi. Interface ek mediator aur intelligent agent ban sakta hai; programming by example action aur language dono.	
Hypertext	**Vannevar Bush ka memex (1945)**; **Nelson (1960s)** ne hypertext (non-linear browsing) ka naam diya; hypermedia, Xanadu project.	1945-60s
Multimodality	**Mode** = insaani communication channel. Kayi channels ek saath (input aur output).	
CSCW	**Computer Supported Cooperative Work**: single user/single computer ka bias khatam; saamaji pehlu. Email sab se kamyab.	
World Wide Web	Hypertext pehle band system tha; **HTTP** aur **HTML** ne publishing aasan ki; critical mass ne maeeshat badal di.	1990s
Agent-based interfaces	Commands se Direct Manipulation, phir **agents** jo proactive aur 'intelligent' hon (avatars, natural language) = language ki taraf wapsi.	
Ubiquitous computing	**Mark Weiser (1991)**: 'sab se gehri technologies woh hain jo gayab ho jati hain.' Computer ko chhota kar ke duniya mein ghol do; aise interactions jo hamari tavajjo na maangein.	1991
Sensor-based aur context-aware	Light, temperature, location, identity ko khud sense kar ke aise interactions jo mahaul ko 'jaante' hon.	

Shneiderman ke Direct Manipulation usool: cheezon ki **visibility**; **incremental action** aur foran feedback; **reversibility** (undo, taake log tajarba karein); har action ki **syntactic correctness** (ghalat command banti hi nahi); **language ki jagah action**.

Paper mein aise likho:
Key HCI paradigms: batch processing, time-sharing, video display units (Sketchpad), programming

toolkits, personal computing, window systems and WIMP, metaphor, direct manipulation (Shneiderman), hypertext, multimodality, CSCW, the World Wide Web, agent-based interfaces, ubiquitous computing (Weiser) and sensor-based, context-aware interaction. Direct manipulation means visibility of objects, incremental action with rapid feedback, reversibility, and replacing language with action.

Chapter 5: Interaction Design Basics

5.1 Design kya hai?

Design ka matlab hai **pabandiyon (constraints) ke andar rehte hue goals hasil karna**. Goals: yeh kis ke liye hai aur wo kyun chahte hain? Constraints: materials, platforms, trade-offs. **Golden rule of design: apne 'materials' ko samjho.** HCI mein materials do hain: **computers** (limitations, capacities, tools) aur **log** (nafsiyati aur saamaji pehlu, insaani ghalti).

Interactions aur interventions: sirf interface nahi, **poori interaction** design karo (misaal: office ka stapler: manual mein likho-print-staple-likho-print-staple; electric mein likho-print-likho-print..-staple: technology ne andaz badal diya). Aur sirf system nahi balke **documentation, manuals, tutorials** bhi.

To err is human: hadsay ki rapat mein aksar 'human error' ko zimmedar thehrate hain. Magar agar concrete ka lintel zyada wazan se toot jaye to hum 'lintel error' nahi kehte, **design error** kehte hain. Insaani ghalti **aam** hai, is liye uske liye design karo. **Central message: the user.**

Paper mein aise likho:
Design is achieving goals within constraints. The golden rule of design is to understand your materials: computers and people. Human error is normal, so systems should be designed for it; users should be treated at least as well as physical materials.

5.2 Design ka process

Process ke qadam: **requirements** (kya hai, kya chahiye), **analysis** (tarteeb aur samajh), **design** (kya karna, kaise faisla), **iteration aur prototyping** (theek karna aur asal zaroorat dhoondna), **implementation aur deployment** (banana aur logon tak pahunchana). Waqt kam hota hai, to **trade-off**: usability, masle dhoondna aur theek karna, kya pehle theek ho. **Ek perfect system bura design hai** (zyada mehnat lagi).

Paper mein aise likho:
The design process includes requirements, analysis, design, iteration and prototyping, and implementation and deployment. Because time is limited, designers must make trade-offs about which usability problems to find and fix.

5.3 User focus: Persona aur Cultural probes

Apne user ko jaano: woh shayad aap jaisa nahi. Un se baat karo, unhein dekho, tasawwur istemal karo.

Tareeqa	Matlab	Example
Persona	Ek **'example' user ki tafseel**. Asli shakhs hona zaroori nahi. Surrogate user ki tarah: 'Betty kya sochegi?' **Tafseel (details) zaroori** hain taake wo asli lage.	Betty, 37 saal, warehouse manager, 5 saal se, do bachche, kam computer training, daayen haath ki harkat thori kam, naye computer system se thori ghabrati hai.

Tareeqa	Matlab	Example
Cultural probes	Jab seedhe observation mushkil ho (ghar, mareez). Logon ko **probe packs** (items jo reaction bhadkayein: deewar sunne ka glass, camera, postcard) dete hain; wo apne mahaul mein record karte hain jo un ke liye ahem hai. Interviews, ideas aur designers ki samajh ke liye.	Ghar mein postcard aur camera dena.

Paper mein aise likho:
A persona is a description of an example user, not necessarily a real person, used as a surrogate user. Details make the persona feel real. Cultural probes are probe packs (for example camera, postcard) given to people so they record what is meaningful to them in their own environment.

5.4 Scenarios (design ki kahaniyan)

Scenario design ke liye ek **kahani** hai: user kya karna chahega, qadam ba qadam. Is mein: **kya nazar aata hai** (sketches/screenshots), **kya karta hai** (keyboard, mouse), **kya soch raha hai**. Fayde: doosron se baat karne (designers, client, users), doosre models ko **validate** karne, dynamics (behavior) dikhane ke liye. **Screenshots = shakal; scenario = behavior.** Scenarios ke saath **play-acting** (mock-up bana kar nakal karna) bhi hota hai.

Linearity: Pros: zindagi aur waqt linear hai, kahani samajhna aasan, concrete hai (ghaltiyan kam). Cons: koi choice ya branch nahi, khaas conditions aur ghair-mutawaqqe cheezein chhoot jati hain. Is liye **kayi scenarios aur kayi methods** use karo.

Paper mein aise likho:
A scenario is a story for design: a step-by-step walkthrough of what users do, see and think. Scenarios help communicate with others, validate other models and express dynamics. Their weakness is linearity: they show one path and miss alternatives, so several scenarios and methods should be used.

5.5 Navigation design

Navigation ka matlab hai system mein **apna raasta dhoondna**. Teen levels: **local** (ek screen se bahar dekhna), **global** (poori site/application ka structure), aur **wider** (doosri applications aur web se rishta). **Levels:** widget choice (menus, buttons), screen design, application navigation design, environment (doosri apps, OS). Web par: elements aur tags (a href), page design, site structure, web aur browser. Physical devices par: controls (buttons, knobs), physical layout, modes of device.

Chaar golden rules of navigation: (1) **Knowing where you are** (2) **Knowing what you can do** (3) **Knowing where you are going / what will happen** (4) **Knowing where you've been / what you've done**.

Breadcrumbs: site ke hierarchy mein aap ka raasta dikhana (Home > Category > Sub-category > Is page), aur live links se upar ke levels par ja sakte ho. **Big button trap:** bade bade buttons jin ke peeche kya hai nahi pata ('other things', 'the thing from outer space'): is se bacho, extra text ki jagah hai to likho. **Modes:** misal: phone ka lock jab jeb mein 'yes' dab jaye to phone book khul jati hai aur 'c' + 'yes' se entry delete: oops! Modes khatarnak ho sakte hain.

Global structure: hierarchical diagrams (application ke hisse, functional separation) aur **network diagrams** (kya kis tak le jata hai, branches, task-oriented). **Deep hierarchy mushkil hai**; Miller ke 7±2 ka ghalat istemal (yeh STM ke liye hai, menu size ke liye nahi); behtar: har screen par zyada items magar screen ke andar structured. **Dialogue:** shaadi ki misaal: general flow (blanks ke saath) hamesha ek, tafseel har dafa alag; computer dialogue = user aur system ke beech interaction ka pattern.

Paper mein aise likho:
Navigation design works at local (single screen), global (whole site or application) and wider (other

applications) levels. The four golden rules of navigation: know where you are, know what you can do, know where you are going or what will happen, and know where you have been or what you have done. Breadcrumbs show the path through a hierarchy.

5.6 Screen design aur layout

Basic usool: grouping, structure, order; **alignment**; **white space**. Pehle poochho user kya kar raha hai? kaun si maloomat, comparisons, tarteeb chahiye? 'Form follows function.' Tools: items ka grouping, tarteeb, decoration (fonts, boxes), alignment, white space.

Usool	Matlab	Note
Grouping aur structure	Jo **logically** saath hain wo **physically** saath rakho. Billing details, Delivery details, Order details alag alag groups.	Form ke groups
Order	Natural tarteeb socho; screen ki tarteeb jawaab ki tarteeb se mile; tab order theek set karo. 'Cake recipe syndrome' se bacho (mix milk and flour, add fruit after beating them: pehle beating batao).	
Decoration	Boxes se logical items group, fonts se emphasis aur headings, magar bohat zyada nahi.	
Alignment (text)	Left align (English): boring magar parhne mein aasan.	
Alignment (names)	Surname se dhoondna ho to surname align karo (Dix, Alan).	
Alignment (numbers)	**Lambi number = bara number**, to decimal points align karo ya integers right-align karo.	
Multiple columns	Gaps ke paar scan karna mushkil; **leaders** (dots), greying ya (ehtiyaat se) 'bad' alignment.	
White space	Sirf 'kya nazar aata hai' nahi, **beech ke faasle** bhi (gaps). Alag karne, structure dene aur highlight karne ke liye.	

Physical controls par bhi yehi usool: grouping (defrost settings, type of food, time), order (1 type of heating, 2 temperature, 3 time, 4 start), decoration (alag functions alag rang, related buttons ke ird gird lines), alignment, white space (gaps).

Paper mein aise likho:
Screen design relies on grouping and structure (logically related items physically together), order that matches the task, decoration, alignment (left-align text, align decimal points for numbers) and white space to separate, structure and highlight items.

5.7 User action aur control: forms aur affordance

Forms aur dialogue boxes: presentation + data input; labels ki alag lambai ka masla; logical layout, task analysis, groups, entry ki natural tarteeb (upar-neeche, left-right, culture ke hisab se), keyboard ke liye **tab order**. **Knowing what to do:** kya active hai, kya passive; kahan click, kahan type; consistent style (web par underlined links); labels aur icons; common actions ke standards; bold = maujooda halat.

Affordance: yeh nafsiyati lafz asal mein **physical objects** ke liye hai: cheez ki **shakal aur size batati hain ke usay kya kar sakte ho**. Chai ke mug ka handle **pakadne (grasping)** ko afford karta hai. Screen par: button jaisi cheez **click** ko afford karti hai, aur icons double-click ko (asli buttons jaisa nahi!). Buttons jaisa dikhne wala 'button' ki tarah kaam karna chahiye.

Paper mein aise likho:
Affordance is a psychological term for physical objects: the shape and size of an object suggest the actions it allows, for example a mug handle affords grasping. On screen, a button-like object affords clicking. Forms need logical grouping, natural order and a correct tab order.

5.8 Information present karna: aesthetics, colour, culture

Purpose matter karta hai: sort order, text ya diagram, scatter graph ya histogram; kagaz ke usool lagao magar **interactivity** bhi jodo (columns ka dobara order karna). **Aesthetics aur utility:** khoobsurat design satisfaction aur productivity barhata hai, magar kabhi khoobsurti aur kaam ka takrao ho sakta hai (text ke peeche background: dekhne mein achha, parhne mein mushkil); mil kar bhi chal sakte hain (consumer products mein key differentiator: iMac).

Colour aur 3D aksar bura use hote hain. Colour: purane monitors ki limited palette; 'kyunke maujood hai' is liye zyada; **colour-blind ka khayal**; kam istemal karo, doosri maloomat ko mazboot karne ke liye. Test: apna monitor sirf grey kar do: kya ab bhi parh sakte ho? 3D: physical maloomat aur kuch graphs ke liye theek, bohat zyada (perspective mein text, 3D pie charts) bura. **Bad colour use:** bina wajah zyada rang, colour blindness, kharab contrast.

Localisation aur internationalisation: alag zubaan aur culture ke liye interface badalna; **globalisation:** aise symbols chuno jo har jagah chalein. Sirf zubaan badalna kaafi nahi (resource database, magar sizes aur left-right order badalte hain); gehre masle: cultural assumptions, symbols ke matlab (tick aur cross: kuch cultures mein +ve/-ve, kuch mein dono 'mark this').

Paper mein aise likho:
Aesthetically pleasing designs increase user satisfaction, but beauty and utility may conflict. Colour and 3D are often misused: use colour sparingly to reinforce other information and consider colour blindness. Localisation changes interfaces for particular cultures and languages; symbols may have different meanings in different cultures.

5.9 Iteration aur prototyping

Pehli dafa mein sahi nahi hota: design -> prototype -> evaluate -> (theek hai? haan to done) warna re-design. Iteration se behtar hote hain aur achhi shuruaat zaroori hai. **Pitfalls:** aahista aahista barhna magar kahan tak? (Malverns ya Matterhorn? yaani chhoti pahari par atak jana, asal ooncha pahaar nahi). Zaroori: **1) achhi starting point** chahiye, **2) samajhna ke ghalat kya hai.**

Paper mein aise likho:
Iteration and prototyping: you never get it right the first time, so design, prototype, evaluate and re-design until the result is acceptable. Pitfalls: a good starting point is needed and the designer must understand what is actually wrong, or the design may only reach a local improvement.

Chapter 6: HCI in the Software Process

6.1 Software lifecycle

Software engineering software banane ke process (**life cycle**) ko samajhne ka maidan hai. **Usability ke liye design har stage par hota hai**, sirf ek alag kaam nahi.

Waterfall model (paani ki aabshar): ek stage poori ho to agli: **Requirements specification -> Architectural design -> Detailed design -> Coding and unit testing -> Integration and testing -> Operation and maintenance.**

Activity	Matlab
Requirements specification	Designer aur customer likhte hain system se kya umeed hai (natural language ya task analysis jaisi sahi zubaan mein).
Architectural design	High-level tafseel: system ko bade components mein baanto aur unka rishta; functional aur non-functional dono requirements poori hon.

Activity	Matlab
Detailed design	Components ko mazeed todna, alag modules ki shinakht; non-functional requirements ke hisaab se.

Verification aur Validation: Verification = 'designing the product **right**' (design specification ke mutabiq hai?). **Validation** = 'designing the **right** product' (asal zaroorat poori karta hai?). **Formality gap:** asal duniya ki requirements aur formal specification ke beech farq; validation hamesha kisi had tak **subjective** saboot par bharosa karti hai.

Interactive systems ka life cycle linear nahi hota; **bohat zyada feedback** hoti hai (har stage se peechay ke stages tak).

Paper mein aise likho:
The waterfall model has requirements specification, architectural design, detailed design, coding and unit testing, integration and testing, and operation and maintenance. Verification means designing the product right; validation means designing the right product. For interactive systems the lifecycle cannot be linear because there is a lot of feedback between stages.

6.2 Usability engineering

Usability ka asal imtihan **user experience ki paimaish** hai. Usability engineering mein **khaas usability measures** requirements ke taur par saaf likhe jate hain. **Usability specification** mein: usability attribute/principle, measuring concept, measuring method, aur **now level, worst case, planned level, best case**.

Example: VCR ka misaal. Attribute: backward recoverability. Measuring concept: ghalat programming sequence ko undo karna. Measuring method: current program undo karne ke explicit user actions ki tadaad. Now level: koi maujooda product undo ki ijazat nahi deta. Worst case: jitne actions se ghalti program ki thi utne. Planned level: zyada se zyada 2 actions. Best case: 1 cancel action.

Masle: usability specification itni tafseel maangti hai jo design ke shuru mein mumkin nahi; spec poori karne se usability lazmi poori nahi hoti.

ISO 9241 usability standard: teen cheezein: **Effectiveness** (jo chahte ho kar sakte ho?), **Efficiency** (bina mehnat zaya kiye?), **Satisfaction** (kya process mein mazaa aaya?).

Usability	Effectiveness measure	Efficiency measure	Satisfaction measure
Suitability for the task	Goals ka percentage achieved	Task complete karne ka time	Satisfaction ki rating scale
Appropriate for trained users	Power features ki tadaad	Expert ke muqable mein relative efficiency	Power features par satisfaction
Learnability	Seekhe gaye functions ka percentage	Criterion tak seekhne ka time	Ease of learning ki rating
Error tolerance	Kamyabi se theek hui ghaltiyon ka percentage	Ghaltiyan theek karne mein waqt	Error handling ki rating

Paper mein aise likho:
Usability engineering makes specific usability measures explicit as requirements. A usability specification gives the attribute, measuring concept, measuring method and the now, worst-case, planned and best-case levels. ISO 9241 defines usability as effectiveness, efficiency and satisfaction.

6.3 Iterative design aur prototyping

Iterative design adhoori requirements ke masle ko door karta hai. **Prototype** intended system ke kuch features ko simulate ya animate karta hai. Qismein: **throw-away** (istemal ke baad phenk do), **incremental** (hissa hissa jorna), **evolutionary** (prototype hi aage jakar asal system banta hai). Management issues: waqt, planning, non-functional features, contracts.

Technique	Matlab
Storyboards	Kahani ki tarah sketches ka silsila; computer par hona zaroori nahi; animated bhi.
Limited functionality simulations	System ka kuch hissa designers khud provide karte hain (HyperCard jaise tools).
Wizard of Oz	User samajhta hai computer jawab de raha hai, asal mein parde ke peeche insaan (wizard) jawab deta hai.

Iterative design ki warning: design inertia (shuru ke bure faislay bure hi rehte hain), aur prototypes mein **asal usability masle** pehchanna, sirf alamaat (symptoms) nahi.

Paper mein aise likho:
Prototypes simulate or animate features of the intended system; types are throw-away, incremental and evolutionary. Techniques include storyboards, limited-functionality simulations and the Wizard of Oz technique. A risk of iterative design is design inertia: early bad decisions stay bad.

6.4 Design rationale

Design rationale woh maloomat hai jo batati hai **system aisa kyun hai** (kaun se faislay, kyun, kaun se options chhore). **Fayde:** life cycle mein communication, design knowledge ka dobara istemal, design discipline, trade-offs ke daleel, design space ka tarteeb, context ki maloomat.

Qisam	Matlab	Example
Process-oriented	Faisla karne ka **silsila (order of deliberation)** mehfooz karta hai.	IBIS / gIBIS
Structure-oriented	Soche gaye design alternatives ki **baad mein banai gayi structure** par zor.	Design space analysis: QOC, DRL

IBIS (Issue-Based Information System): main elements: **issues** (hierarchy, ek root issue), **positions** (issue ke mumkin hal), **arguments** (positions aur issues ke rishte ko badalte hain). gIBIS graphical version hai. **QOC (Design Space Analysis): Questions** (design ke bare issues), **Options** (alternative hal), **Criteria** (options ko parakhne ke maiyar). DRL, QOC jaisa magar zyada formal zubaan ke saath.

Psychological design rationale: task-artefact cycle ko support karta hai (user ke kaam un systems se badalte hain jo wo use karte hain); designers tasks pehchante hain, scenarios banate hain, users ko observe karte hain, **psychological claims** saaf likhte hain; bure pehlu agle iteration ko behtar banate hain.

Paper mein aise likho:
Design rationale is information that explains why a system is the way it is. Benefits: communication across the life cycle, reuse of design knowledge, design discipline, recording of trade-offs. Process-oriented rationale (IBIS: issues, positions, arguments) keeps the order of decisions; structure-oriented rationale (QOC: questions, options, criteria) structures the considered alternatives.

Chapter 7: Design Rules

7.1 Design rules ki qismein

Maqsad: **zyada se zyada usability**. Teen cheezein: **Principles** (abstract design rules; low authority, high generality), **Standards** (khaas rules; high authority, limited application), **Guidelines** (kam authority, zyada aam application). Aur **design patterns** design ka tajurba mehfooz aur dobara istemal karne ke liye.

Qisam	Matlab	Example
Principles	Abstract, aam; authority kam, generality zyada.	Learnability, flexibility, robustness
Guidelines	Suggestive; kam authority, zyada aam; textbooks aur style guides.	Style guide
Standards	Qaumi/aalmi idaron ke; **high authority**, low level of detail, hardware standards zyada.	ISO 9241

Paper mein aise likho:
Design rules are principles (abstract, low authority, high generality), guidelines (more specific, lower authority, wider application) and standards (specific, high authority, limited application). Standards need a sound underlying theory and slowly changing technology; ISO 9241 defines usability as effectiveness, efficiency and satisfaction.

7.2 Usability ke principles: Learnability, Flexibility, Robustness

Teen bare hisse. Yaad rakhne ka asaan tareeqa: **L-F-R**.

1) Learnability: naye users kitni **aasani se** kaam shuru kar ke behtareen performance tak pahunchte hain.

Principle	Matlab
Predictability	Past interaction se **aaindah actions ka asar** tay karna. (operation visibility)
Synthesizability	**Guzre actions ka asar** samajhna. (immediate vs eventual honesty)
Familiarity	Purani maloomat naye system par kaise lagu hon. (guessability, affordance)
Generalizability	Khaas interaction ki maloomat naye halaat tak phailana.
Consistency	Milte julte halaat mein input/output behavior ka ek jaisa hona.

2) Flexibility: user aur system ke maloomat ke aadan-pradan ke **kayi tareeqay**.

Principle	Matlab
Dialogue initiative	System ki lagayi pabandiyon se azadi. (system vs user pre-emptiveness)
Multithreading	Ek se zyada task ek waqt mein. (concurrent vs interleaving; multimodality)
Task migratability	Kaam ki zimmedari user aur system ke beech badalna. (jaise spell-check)
Substitutivity	Barabar values ko aapas mein badalne ki ijazat. (representation multiplicity; equal opportunity)
Customizability	Interface ko user (adaptability) ya system (adaptivity) badal sake.

3) Robustness: goal ke hasil honay ko tay karne aur parakhne mein user ko dee gayi **madad ka level**.

Principle	Matlab
Observability	System ki andar ki halat ko uski nazar aane wali shakal se parakhna. (browsability, defaults, reachability, persistence, operation visibility)
Recoverability	Ghalti pehchan kar **sudharna**. (reachability, forward/backward recovery, commensurate effort)
Responsiveness	User ko system ki raftar kaisi lagti hai.
Task conformance	System ki services user ke saare tasks ko kitna support karti hain. (task completeness, task adequacy)

Paper mein aise likho:
Principles to support usability: learnability (predictability, synthesizability, familiarity, generalizability, consistency), flexibility (dialogue initiative, multithreading, task migratability, substitutivity, customizability) and robustness (observability, recoverability, responsiveness, task conformance).

7.3 Golden rules aur heuristics

Yeh 'broad brush' rules hain: achhe design ki checklist. **Kuch na hone se behtar hai ke yeh hon.** Mashhoor collections: **Nielsen ke 10 heuristics** (chapter 9), **Shneiderman ke 8 Golden Rules**, aur **Norman ke 7 Principles**.

#	Shneiderman ke 8 Golden Rules
1	Strive for consistency (ek jaisa pan)
2	Enable frequent users to use shortcuts
3	Offer informative feedback
4	Design dialogs to yield closure (kaam ke aakhir mein ehsaas ke mukammal hua)
5	Offer error prevention and simple error handling
6	Permit easy reversal of actions (undo)
7	Support internal locus of control (user control mein ho)
8	Reduce short-term memory load

#	Norman ke 7 Principles
1	Use both knowledge in the world and knowledge in the head.
2	Simplify the structure of tasks.
3	Make things visible: bridge the gulfs of Execution and Evaluation.
4	Get the mappings right.
5	Exploit the power of constraints, both natural and artificial.
6	Design for error.
7	When all else fails, standardize.

Paper mein aise likho:
Shneiderman's eight golden rules: strive for consistency, enable shortcuts for frequent users, offer informative feedback, design dialogs to yield closure, offer error prevention and simple error handling, permit easy reversal of actions, support internal locus of control, and reduce short-term memory load.

7.4 HCI design patterns

Pattern ek **aazmaya hua, baar baar aane wale masle ka invariant hal** hai ek khaas context mein. Yeh **architecture** (Christopher Alexander) se aaya: 'har kamray mein do taraf se roshni'. HCI mein misal: **'Go back to a safe place'**. Patterns akele nahi hote, aapas mein links se ek **pattern language** banate hain jis se poora design ban sakta hai.

Khasiyaten: design practice mehfooz karte hain (nazariya nahi); achhe examples ki aam khasiyat; alag levels par (saamaji, tanzeemi, conceptual, tafseeli); insani values; saaf aur parhne mein aasan taake sab stakeholders baat kar saken; pattern language 'generative' ho.

Paper mein aise likho:
A design pattern is an invariant solution to a recurrent problem within a specific context. Patterns capture design practice, represent knowledge at varying levels, are easy to read for all stakeholders and, linked together in a pattern language, help generate complete designs. Example: 'Go back to a safe place'.

Chapter 8: Implementation Support

8.1 Programming tools ke levels

Interface banane ke liye programmer ko **teen levels ke tools** milte hain (jitna upar, utna aasan): **Windowing systems** (alag aur ek saath user-system activity ki buniyaad), **Interaction toolkits** (programming ko user ke nazariye ke qareeb laate hain), **User Interface Management Systems (UIMS)** (presentation aur functionality ka rishta control karte hain). Pehle coding hardware-specific aur interaction-technique-specific thi; ab nahi.

Paper mein aise likho:
Programming support has three layers: windowing systems (core support for separate and simultaneous user-system activity), interaction toolkits (bring programming closer to the user's perception) and user interface management systems (control the relationship between presentation and functionality).

8.2 Windowing systems

Device independence: programmer 'abstract terminal' ke liye likhta hai, device drivers ke liye nahi; **image models** (pixels, PostScript, GKS, PHIGS). **Resource sharing:** user ke kayi kaam ek saath, har application alag (isolation).

Teen software architectures (device driver sab mein alag hota hai): (1) har application khud sab processes sambhale: synchronization ki fikr, **portability kam**. (2) management OS ke **kernel** mein: applications OS se bandhi. (3) management ek **alag application** ke taur par: **sab se zyada portability**.

X Windows architecture (client-server): pixel imaging model aur pointing; **X protocol** server-client baat cheet tay karta hai; alag **window manager client** policies lagata hai: input focus kaise badle, tiled ya overlapping windows, clients ke beech data transfer.

Paper mein aise likho:
A windowing system provides device independence (an abstract terminal and image models such as pixels or PostScript) and resource sharing among independent processes. Three architectures exist: each application manages everything, management inside the OS kernel, or management as a separate application (most portable). X Windows uses a client-server architecture with a separate window manager.

8.3 Application programming: event loop aur notification

Application kaise likhte hain, iske do tareeqay:

Tareeqa	Matlab	Aasani / mushkil
Read-evaluation loop (event loop)	Programme ek loop mein **event parhta** hai aur uski type ke hisaab se case chalata hai: *repeat read-event; case event.type...* Application loop ko control karta hai.	Modal dialog aasaan (bas ek aur read-event loop); non-modal mushkil (bohat pechida main loop).
Notification-based	Aap batate ho kaun si **event par kaun sa function chale** (menu.setAction('Save', mySave)); system khud us ko call karta hai (callback).	Non-modal dialog aasaan (bas ek handler); modal mushkil (bohat mode flags).

'Going with the grain': system ka style interfaces par asar daalta hai. **Khabardar:** agar aap explicitly design na karo to wohi ban jata hai jo implementation aasaan banaye; **implementation ko design nahi chalana chahiye**.

Paper mein aise likho:
In the read-evaluation (event) loop style, the program loops reading events and handling them by type. In the notification-based style, the programmer registers callbacks for events and the system calls them. A modal dialog is easy with an event loop and hard with notification; a non-modal dialog is the opposite. Implementation should not drive design.

8.4 Toolkits

Interaction objects (widgets/gadgets) input aur output ko jorte hain (jaise button: dikhna aur dabana). Toolkits is level ki **abstraction** dete hain. Fayde: **consistency** aur **generalizability** (ek jaisa look and feel), aur object-oriented programming ke liye mauzoon. **Java mein: AWT** (Abstract Windowing Toolkit): buttons, menus waghera ki classes; notification based; AWT 1.0 mein widgets ko subclass karna parta tha, 1.1 ke baad callback objects. **Swing:** AWT ke upar, zyada high-level features, **MVC architecture**.

Paper mein aise likho:
Toolkits provide interaction objects (widgets) that link input and output, promoting consistency and generalizability through a similar look and feel. Java's AWT is notification-based and Swing is built on AWT using the MVC architecture.

8.5 UIMS (User Interface Management Systems)

Toolkits non-programmers ke liye mushkil hain, to UIMS ek level upar. Concerns: **conceptual architecture**, **implementation techniques**, **support infrastructure**. Aur naam: UI development system (UIDS), UI development environment (UIDE), jaise **Visual Basic**. **UIMS ek conceptual architecture ke taur par:** application semantics aur presentation ko **alag** rakhna. Is se: **portability**, **reusability**, **multiple interfaces** (ek hi functionality ke kayi interfaces), **customizability**.

Paper mein aise likho:
A UIMS adds a level above toolkits. As a conceptual architecture it separates application semantics from presentation, which improves portability, reusability, support for multiple interfaces and customizability. Examples of UI development environments include Visual Basic.

8.6 Seeheim model aur Arch/Slinky

Seeheim model interface ko teen logical hisson mein baanta hai: **Presentation** (lexical: jismani input/output), **Dialogue control** (syntactic: kaun si cheez kis ke baad), aur **Functionality / application interface** (semantic: application ka asal kaam). Neeche **switch** (bypass): rapid semantic feedback ke liye presentation aur application ke beech seedha rasta, jo dialogue control se regulate hota hai.

Semantic feedback: teen qisam ka feedback: **lexical** (mouse ki harkat), **syntactic** (menu highlight), **semantic** (numbers ka sum badal gaya). Semantic feedback aksar dheema hota hai. Misal: freehand

drawing, ya file drag karte waqt trash can highlight ho.

Arch/Slinky: Seeheim se zyada layers; lexical aur physical ko alag kiya. 'Slinky' spring ki tarah alag layers alag systems mein mote (zyada ahem) ya patle ho sakte hain.

Paper mein aise likho:
The Seeheim model divides an interface into presentation (lexical), dialogue control (syntactic) and functionality (semantic), with a switch/bypass for rapid semantic feedback. Arch/Slinky adds more layers and distinguishes lexical from physical; layers may be thicker or thinner in different systems.

8.7 MVC aur PAC

Seeheim ke components bade hote hain; chhote components aksar aasan hote hain (OO toolkits ke saath). **MVC (Model-View-Controller)** Smalltalk mein istemal hua:

Hissa	Kaam
Model	Component ki **andar ki logical halat**.
View	Wo screen par **kaise nazar aata hai** (rendering).
Controller	**User ke input ko process** karta hai.

MVC ka masla: asal mein yeh pipeline lagta hai (input -> controller -> model -> view -> output), magar graphical interface mein input ka matlab output ke hawale se hota hai (mouse click ka matlab usay pata chale ke kya click hua, aur wo to **view** jaanta hai). To amal mein controller view se baat karta hai: **alag-alag hona mukammal nahi**.

PAC (Presentation-Abstraction-Control): Seeheim ke qareeb. **Abstraction** (component ki logical halat), **Presentation** (input/output sambhalta hai), **Control** (in dono ke beech vasta; hierarchy aur kayi views manage karta hai). PAC kayi lihaz se saaf hai magar **MVC amal mein zyada istemal hota hai** (jaise Java Swing).

Paper mein aise likho:
MVC (Model-View-Controller) separates the internal logical state (model), how it is rendered (view) and input processing (controller). In practice the controller talks to the view, so the separation is incomplete. PAC (Presentation-Abstraction-Control) is closer to Seeheim: abstraction holds the logical state, presentation handles input/output and control mediates between them, managing hierarchy and multiple views.

8.8 Dialogue controller ke techniques aur graphical specification

UIMS mein dialogue controller ke tareeqe: **menu networks, state transition diagrams, grammar notations, event languages, declarative languages, constraints, graphical specification**. **Constraints:** 'kya hota hai' likhne ki jagah likho 'kya sach hona chahiye' (groupware mein bhi). **Graphical specification:** screen par components draw karo aur actions script ya program se jodo: bohat mashhoor (**Visual Basic, Dreamweaver, Flash**). Nuqsan: poore system ke raaste 'dekhna' mushkil, sirf ek screen par focus.

Dialogue control ka drift: internal control (read-evaluation loop) -> **external control** (application semantics aur presentation se azad) -> **presentation control** (graphical specification).

Paper mein aise likho:
Dialogue controller techniques include menu networks, state transition diagrams, grammar notations, event languages, declarative languages, constraints and graphical specification. Graphical specification (Visual Basic, Dreamweaver, Flash) lets the designer draw components and attach actions, but it is hard to see the paths through the whole system.

Chapter 9: Evaluation Techniques

9.1 Evaluation kya hai?

Evaluation system ki **usability aur functionality** ko parakhna hai. Yeh **lab, field ya users ke saath** hota hai, **design aur implementation** dono ko parakhta hai, aur design life cycle ke **har stage par** hona chahiye. **3 maqasid:** system ki functionality ki had parakhna; interface ka user par asar parakhna; khaas masle pehchanna.

Paper mein aise likho:
Evaluation tests the usability and functionality of a system and should be done at all stages of the design life cycle. Its goals are to assess the extent of system functionality, assess the effect of the interface on the user, and identify specific problems.

9.2 Designs ko evaluate karna (users ke baghair)

Tareeqa	Matlab	Example
Cognitive Walkthrough	**Polson et al.** Parakhta hai ke design user ko task seekhne mein kitna madad deta hai. Aam taur par cognitive psychology ka expert design par 'chal kar' dekhta hai aur psychological usoolon se masle dhoondta hai; forms rehnumai karte hain. Har task par: interaction ka user par kya asar? kaun se cognitive processes chahiye? seekhne mein kya masle? Focus: kya design user ko **sahi goals** banane par laata hai?	Expert 'Photo email karna' task par qadam ba qadam jaata hai.
Heuristic Evaluation	**Nielsen aur Molich.** Usability criteria (heuristics) tay; experts dekhte hain ke design in ki **khilaf-warzi** karta hai ya nahi. Misal heuristics: system ka behavior predictable, consistent, feedback mile. Design ko **'debug'** karta hai.	Teen experts ek app ko 10 heuristics par check karte hain.
Review-based evaluation	Pehle ki research ke nateeje design ke hissay ko support ya reject karne ke liye. Dhyan: kya nateeje naye design par lagu hote hain?	Fitts' law se button size parakhna.
Model-based evaluation	Cognitive models se design options ko **chhaantna**; jaise **GOMS** user ki performance ki pesh-goi. Design rationale bhi madadgar.	

Paper mein aise likho:
Cognitive walkthrough (Polson et al.) evaluates how well a design supports the user in learning a task, by an expert walking through it. Heuristic evaluation (Nielsen and Molich) has experts check whether a design violates agreed usability heuristics. Review-based evaluation uses results from the literature, and model-based evaluation uses cognitive models such as GOMS.

9.3 Users ke saath evaluation: Laboratory vs Field

Qisam	Fayde	Nuqsan	Kab munasib
Laboratory studies	Khaas equipment maujood; **be-rukawat mahaul**.	**Context ka na hona**; kayi users ke cooperation ko dekhna mushkil.	Jab system ki jagah khatarnak ya mushkil ho; ek user ke controlled experiment ke liye.
Field studies	**Natural mahaul**; context bacha rehta hai (observation se badal sakta hai); lambay arse ke mutale mumkin.	Distraction, shor.	Jab context zaroori ho; lambay arse ke mutale.

Paper mein aise likho:
Laboratory studies offer specialist equipment and an uninterrupted environment but lack context. Field

studies take place in the natural environment, keep context and allow longitudinal studies, but suffer distractions and noise.

9.4 Experimental evaluation

Controlled evaluation jis mein interaction ke kisi khaas pehlu ko parakha jata hai. Evaluator ek **hypothesis** chunta hai, aur kayi **experimental conditions** jo sirf ek controlled variable ki value mein alag hon; behavior ke naap mein tabdeeli ko conditions ki wajah se mana jata hai.

Factor	Matlab
Subjects	**Kaun**: numainda aur kaafi tadaad ke participants.
Variables	**Independent variable (IV):** jo characteristic badli jaye taake conditions bane (jaise interface style, menu items ki tadaad). **Dependent variable (DV):** jo measure ho (time taken, errors ki tadaad).
Hypothesis	**Jo sabit karna hai**: IV aur DV ke hawale se prediction. Misal: 'font size kam hone se error rate barhega.' **Null hypothesis:** 'conditions mein koi farq nahi' (font size se kuch nahi badla); maqsad usay **ghalat sabit** karna.
Experimental design	**Kaise** karna hai.

Experimental design ki do qismein: Within-groups: har subject har condition mein. Transfer of learning ho sakta hai; sasta, user variation ka kam asar. **Between-groups:** har subject sirf ek condition mein. Transfer of learning nahi; zyada users chahiye; variation se nateeje bias ho sakte hain.

Data ka analysis: statistics se pehle data ko **dekho**, **original data save** karo. Technique ka intikhab data ki qisam par: **discrete** (limited values) ya **continuous** (koi bhi value). **Parametric tests:** normal distribution maanti hain, robust aur powerful. **Non-parametric:** normal distribution nahi maanti, kam powerful, zyada reliable. **Contingency table:** data ko discrete attributes se classify kar ke har group ki ginti. Zaroori sawal: kya farq hai? kitna bara? andaza kitna sahi?

Groups par experiments mushkil hain: subject groups (zyada logon ka kharcha, settle hone mein waqt, zyada variation, timetable mushkil), task (cooperation hona chahiye: creative task jaise report likhna, decision game jaise desert survival, control task jaise ARKola bottling plant), data gathering (kayi cameras + logging: synchronisation aur volume ka masla), analysis (groups mein bohat variation: within-groups experiments, micro-analysis, qualitative). **Field studies** zyada realistic: 'distributed cognition', asal action **situated action** hai, physical aur saamaji mahaul dono ahem.

Paper mein aise likho:
Experimental evaluation tests a hypothesis by comparing conditions that differ only in a controlled variable. The independent variable is what is changed; the dependent variable is what is measured. The null hypothesis states there is no difference between conditions and the aim is to disprove it. In a within-groups design each subject does every condition; in a between-groups design each subject does only one.

9.5 Observational methods

Tareeqa	Matlab	Fayde / nuqsan
Think Aloud	User task karte hue **bolta rehta hai** ke kya aur kyun kar raha hai, kya samajh raha hai.	Fayde: aasaan, kam maharat chahiye, asal istemal dikhata hai. Nuqsan: subjective, selective, bolna kaam badal sakta hai.
Cooperative evaluation	Think aloud ka variation: user aur evaluator **milkar** evaluate karte hain, dono ek doosre se sawal pooch sakte hain.	Zyada fayde: kam pabandiyan, user ko tanqeed ki hosla afzai, wazahat mumkin.
Protocol analysis	Record karne ke tareeqe: paper-pencil (sasta, likhne ki raftar tak), audio (think aloud ke liye, doosre protocols se milana mushkil), video (sahi aur realistic, khaas equipment, intrusive), computer logging (khud ba khud, unobtrusive, data zyada), user notebooks (coarse magar lambay mutale ke liye).	
Post-task walkthrough	Task ke **baad** user ko transcript dikha kar us se pooch-gach. Iraade bharne ke liye.	Fayde: analyst ko waqt, kam rukawat. Nuqsan: taazgi nahi, baad ki tashreeh.

Paper mein aise likho:
Observational methods: think aloud (the user describes what they are doing and why), cooperative evaluation (user and evaluator collaborate and can ask each other questions), protocol analysis (paper, audio, video, computer logging, user notebooks) and post-task walkthroughs (the user comments on a recording after the task).

9.6 Query techniques aur Physiological methods

Tareeqa	Matlab	Fayde / nuqsan
Interviews	Analyst user se **ek ek karke** sawal karta hai, aam taur par tayyar shuda sawal. Informal, subjective, sasta.	Fayde: context ke mutabiq badal sakte hain, issues gehrai se, ghair-mutawaqqe masle. Nuqsan: subjective, waqt talab.
Questionnaires	**Fixed sawalon** ka set. Fayde: jaldi, bare group tak, rigor se analyze. Nuqsan: kam flexible, kam probing. Sawal ki qismein: **general, open-ended, scalar, multi-choice, ranked**.	
Eye tracking	Sar ya desk par lagi equipment aankh ki position track karti hai; aankh ki harkat display ke cognitive processing ko dikhati hai. **Fixations** (theher kar dekhna; tadaad aur muddat mushkil ka andaza), **saccades** (tezi se chhalang), **scan paths** (seedha target par chhoti fixation behtareen).	
Physiological measurement	Emotional response jismani tabdeeli se juri: **dil ki activity** (blood pressure, pulse), **sweat glands: GSR**, **muscle: EMG**, **dimagh: EEG**. In ki tashreeh mushkil, research chahiye.	

Paper mein aise likho:
Query techniques: interviews (one-to-one, flexible but subjective and time consuming) and questionnaires (fixed questions, quick and reach many users but less flexible). Eye tracking measures fixations, saccades and scan paths. Physiological measures include heart activity, galvanic skin response (GSR),

EMG and EEG, but they are difficult to interpret.

9.7 Evaluation method kaise chunein?

Chunav in factors par: **kab** (design ya implementation), **style** (laboratory ya field), **kitna objective** (subjective ya objective), **measures ki qisam** (qualitative ya quantitative), **maloomat ka level** (high-level ya low-level), **interference ka level** (obtrusive ya unobtrusive), aur **resources** (waqt, subjects, equipment, expertise). Evaluating implementations ke liye ek **artefact** chahiye: simulation, prototype ya poora implementation.

Paper mein aise likho:
Choosing an evaluation method depends on when in the process (design or implementation), laboratory or field, how objective, qualitative or quantitative measures, level of information, level of interference, and the available resources (time, subjects, equipment, expertise).

Aakhir mein: Quick Revision

Chapter	Yaad rakhne wali cheezein
Ch1 Human	Vision (rods/cones, colour, reading), hearing (pitch, loudness, timbre), touch, Fitts' law; memory: sensory/STM (7±2)/LTM (episodic, semantic; frames, scripts, production rules); deduction/induction/abduction; slip vs mistake; emotion.
Ch2 Computer	Input (keyboard, mouse, touch...), output (CRT vs LCD), VR, printing/fonts/OCR, RAM vs disks, virtual memory, compression, Moore's law, networks.
Ch3 Interaction	Domain/goal/task; Norman 7 stages; gulf of execution/evaluation; Abowd-Beale (U, I, S, O); ergonomics; interaction styles; WIMP; value.
Ch4 Paradigms	Batch, time-sharing, Sketchpad, Engelbart, personal computing, WIMP, metaphor, direct manipulation, hypertext, CSCW, WWW, agents, ubiquitous computing.
Ch5 Design basics	Persona, scenarios, navigation (4 golden rules, breadcrumbs), screen design, affordance, colour, iteration.
Ch6 Software process	Waterfall, V&V;, usability engineering, ISO 9241 (effectiveness, efficiency, satisfaction), prototyping, design rationale (IBIS, QOC).
Ch7 Design rules	Learnability, flexibility, robustness; Shneiderman 8; Norman 7; patterns.
Ch8 Implementation	Windowing systems, event loop vs notification, toolkits, UIMS, Seeheim, MVC, PAC.
Ch9 Evaluation	Walkthrough, heuristic evaluation, lab vs field, experiments (IV, DV, hypothesis), think aloud, interviews, questionnaires, eye tracking.

Formulas aur numbers jo yaad rakhne hain: Fitts' law: Mt = a + b log2(D/S + 1). STM: **7 ± 2 chunks**. Reaction time: visual ~200 ms, auditory ~150 ms, pain ~700 ms. Colour blindness: **8% men, 1% women**. Hearing range: 20 Hz - 15 kHz. Moore's law: processor ~18 mahine mein double, memory ~12 mahine.

Paper tips: (1) 'Define' wale sawal mein pehle definition phir example. (2) 'Difference' mein **table** banao. (3) Lists (jaise Shneiderman 8, Norman 7, Norman 7 stages) **number laga kar** likho. (4) Diagram wale topics (Norman model, Abowd-Beale, Seeheim, MVC) mein chhota sketch zaroor banao. Allah kare paper behtareen jaye, Moto!